

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 7: Ethics and Ownership

Subtopic 7.1: Ethics and Ownership

Concept 7.1.1: Professional Ethics and Bodies (BCS, IEEE)

Total Questions: 10

Mark Schemes begin on Page 6

Question 1 | Source: 9618_w25_qp_11 (Q8.b)

8 One ethical consideration for a student connecting their personal computer to the school network is the risk of spreading malware on the network.

(b) Give **three** other ethical considerations for a student using their personal computer to connect to the school network.

1
.....
2
.....
3
.....

[3]

Question 2 | Source: 9618_w25_qp_12 (Q2.b)

(b) Describe the purpose of a code of conduct.

.....
.....
.....
.....

[2]

Question 3 | Source: 9618_s25_qp_13 (Q3.d)

3 A programmer is writing a program in a high-level language.

(d) Describe the possible consequences of the programmer **not** joining a professional ethical body.

.....
.....
.....
.....
.....
.....

[3]

Question 4 | Source: 9618_w24_qp_12 (Q5.a)

5 A computer programmer is learning about the importance of acting ethically at work.

- (a) Explain the reasons why the computer programmer needs to act ethically towards colleagues and the public.

Colleagues

.....

.....

.....

The public

.....

.....

.....

[4]

Question 5 | Source: 9618_w24_qp_13 (Q5.b)

- 5 A computer programmer writes programs that are distributed with a software licence.

- (b) Explain the reasons why it is important for the computer programmer to join a professional ethical body.

.....

.....

.....

.....

.....

.....

[3]

Question 6 | Source: 9618_w23_qp_13 (Q6.b)

- (b) Explain why it is important for a programmer to join a professional ethical body.

.....

.....

.....

.....

.....

.....

.....

.....

[4]

Question 7 | Source: 9618_w21_qp_11 (Q4.a)

4 Francis is starting his first job as a software developer for a multinational company.

(a) Francis has been advised to join a professional ethical body.

Describe the benefits to Francis of joining a professional ethical body.

.....

.....

.....

.....

.....

..... [3]

Question 8 | Source: 9618_w21_qp_11 (Q4.b.i)

4 Francis is starting his first job as a software developer for a multinational company.

(b) Francis is shown the software he will be working on. He is unfamiliar with the Integrated Development Environment (IDE) he is required to use.

(i) Describe the ways in which Francis can act ethically in this situation.

.....

.....

.....

..... [2]

Question 9 | Source: 9618_w21_qp_11 (Q4.c)

4 Francis is starting his first job as a software developer for a multinational company.

(c) Francis is part of a team writing a program. He finds an error in part of the program that has already been tested. He decides not to tell anyone because he is worried about the consequences.

Explain the reasons why Francis acted unethically in this situation.

.....

.....

.....

..... [2]

Question 10 | Source: 9618_s21_qp_12 (Q2.a)

2 Aisha manages a team of software developers.

(a) Explain the reasons why it is important that Aisha acts ethically in relation to her team members.

.....

.....

.....

..... [2]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_11 (Q8.b)

8(b)	1 mark per bullet point, max 3 marks e. g. • Students should consider privacy issues • Students should not make inappropriate use of social networking // Students should not engage in cyberbullying • Students should not copy work that is not their own • Students should abide by the school rules/guidelines for accessing the network • Students should only use the school network for legitimate purposes • Students should not hack / crack other computers	3
------	---	----------

Mark Scheme for Question 2 | Source: 9618_w25_ms_12 (Q2.b)

2(b)	1 mark per bullet point, max 2 marks e.g. • to create a safe, respectable and professional working environment for all employees • to ensure that employees understand the expectations • ... and the consequences of their actions • to provide a clear outline of what is considered good practice • so there are clear rules to follow in certain situations • to protect the reputation of the organisation // maintain public trust	2
------	--	----------

Mark Scheme for Question 3 | Source: 9618_s25_ms_13 (Q3.d)

3(d)	1 mark each to max 3 e.g. • There is no / limited access to legal advice • in case action is taken against them • There are fewer networking opportunities • so they could miss out on contacts / jobs • There would be less access to training • There would be no clear laid out ethical guidelines • and / or people to discuss potential ethical problems with • possibly leading to inappropriate / unethical actions • which might lead to legal proceedings / recourse	3
------	--	----------

Mark Scheme for Question 4 | Source: 9618_w24_ms_12 (Q5.a)

5(a)	1 mark for each bullet point max 2 for public max 2 for colleagues e.g. Colleagues: The programmer should: • Treat colleagues fairly and avoid acts of discrimination • Be prepared to accept / offer critique • ... so that technical work can be improved • Credit the contributions of colleagues • ... so that they feel valued / respected • Help / train colleagues etc. e.g. The public: The programmer should: • Maintain the health, safety and welfare of the public • ... so as not to endanger anyone • Be honest / realistic in making claims about the software • ... so that public trust is maintained • Maintain the security of the public's data etc.	4
------	---	---

Mark Scheme for Question 5 | Source: 9618_w24_ms_13 (Q5.b)

5(b)	1 mark for each bullet point (max 3) e.g. • The professional ethical body has ethical guidelines to follow • ... so clients/other staff know the standards being applied • ... so the computer programmer does not have to decide what is ethical it's written down • Clients/staff know the computer programmer is reputable • ... there may be a test/requirement for entry • ... giving recognition of the computer programmer's skills/knowledge • The professional ethical body provides help and support • ... for example, if the computer programmer needs legal advice • The professional ethical body runs training courses • ... to keep the computer programmer's skills up to date	3
------	--	---

Mark Scheme for Question 6 | Source: 9618_w23_ms_13 (Q6.b)

6(b)	1 mark for each bullet point (max 4). • The programmer has ethical guidelines to follow • ...so purchasers of programs know the standards being applied • ...so programmer does not have to decide what is ethical or not • Clients know programmer is reputable • ...recognition of programmer's skills/knowledge • There may be an entry requirement / exam • ... so clients know that the programmer is competent • Professional ethical body provides help and support • Such as legal advice // by appropriate example • Enables programmer to attend the ethical body's training courses • ...to keep skills up to date // to develop skills	4
------	---	----------

Mark Scheme for Question 7 | Source: 9618_w21_ms_11 (Q4.a)

4(a)	1 mark per bullet point to max 3 e.g. • He has ethical guidelines to follow • ... so clients/other staff know the standards being applied • ... so he does not have to decide what is ethical it's written down • Clients / staff know he is reputable • ... recognition of his skills / knowledge • ... there may be a test / requirements for entry • They provide help and support • ... for example if he needs legal advice • They run training courses • ... to keep his skills up-to-date	3
------	--	----------

Mark Scheme for Question 8 | Source: 9618_w21_ms_11 (Q4.b.i)

4(b)(i)	1 mark per bullet point to max 2 e.g. • He can tell the manager he has not used it • ... and how he will get up-to-date • He can perform his own research on how to use it • He can explain to the manager that he needs additional training • He can(ask the manager to book on a training course • He can ask for a mentor / to shadow someone • He can practice at home before starting	2
---------	---	----------

Mark Scheme for Question 9 | Source: 9618_w21_ms_11 (Q4.c)

4(c)	1 mark per bullet point to max 2 e.g. • He didn't act in best interest of product • ... because the product might fail because he didn't report the error • He didn't act in best interest of client • ... because if the product does not work then they have been let down because he didn't report the error • He didn't act in the best interest of the profession • ... he is letting his profession down because he didn't report the error • He didn't act in best interest of the company • ... not correcting the error early could lead to later problems	2
------	---	----------

Mark Scheme for Question 10 | Source: 9618_s21_ms_12 (Q2.a)

2(a)	1 mark per bullet point to max 2 • To make sure the team members feel valued • To get the best work out of the team • To enable them to work well together • To enable them to create the best product for the client	2
------	--	----------